

Project Workload Distribution

Campus Connect - Real-Time Client-Server Text Chat Application

Team Members: Marwan, Karim, Mohamed, Sara

Project Requirements Covered

The project implements a real-time client-server text chat application with a Qt graphical client, a command-line Boost.Asio server, TCP socket networking, JSON message serialization/deserialization, asynchronous I/O, separated GUI/business/networking layers, and automated unit testing.

Marwan

- Networking requirement: Worked on client/server networking behavior and helped connect the Qt client to the server using TCP sockets.
- Server-side requirement: Helped implement the Boost.Asio CLI server, including asynchronous connection handling and multi-client communication.
- JSON communication requirement: Contributed to message routing and session handling for structured JSON messages between clients and server.
- Bonus features: Implemented the bonus functionality, including multiple users/private chats, chat groups, and media/attachment support.

Karim

- GUI requirement: Worked on Qt Designer integration and interface polishing for the application screens.
- User interaction requirement: Helped improve the forms, buttons, and visible user feedback used by the client application.
- Networking validation requirement: Helped validate communication between clients and the server during testing.
- Integration requirement: Helped ensure the GUI worked smoothly with the controller/business logic layer.

Mohamed

- GUI requirement: Worked on Qt Designer integration and interface polishing for the client-side views.
- Testing requirement: Worked on integration testing and helped verify that client/server communication behaved correctly.
- Final system integration requirement: Contributed to final system testing, debugging, and checking that the app works as one complete system.
- Reliability requirement: Helped identify and debug issues that affected the final demo flow.

Sara

- GUI requirement: Worked on Qt Designer integration and interface polishing for the application screens.
- Networking validation requirement: Helped validate communication between multiple clients and the server.
- Presentation/demo requirement: Worked on the presentation slides and helped prepare the project explanation for the final discussion.

- Documentation requirement: Helped organize project material for the final delivery.

Bonus Features Implemented by Marwan

- Multiple users and private chats: Marwan implemented the bonus functionality for multiple connected users, displaying online users, and allowing private chat with another user.
- Chat groups: Marwan implemented the bonus functionality for creating/joining groups and sending group messages.
- Media and attachments: Marwan implemented extra media and attachment functionality, which strengthens the final product beyond a basic text-only chat.

Summary

Overall, the team contributed to the required GUI, networking, JSON protocol, server implementation, separation of concerns, testing, integration, and final presentation.

Marwan also implemented the bonus-level features: multiple users/private chats, chat groups, and media/attachment support.